National University of Computer and Emerging Sciences



Lab Manual 04
Computer Organization
and Assembly Language

Course Instructor	Mr. Salman Mubarik
Lab Instructor	Muhammad Hasan
Lab Instructor Email ID	m.hasan@lhr.nu.edu.pk
Semester	Fall 2026

FAST - School of Computing
FAST-NU, Lahore Campus

Contents

Objectives: 3

Activities: 3

Activity 1 (Loop and Array Processing Tasks): 3

Activity 2 (Bit Manipulation and Shifting Tasks): 3

Objectives:

- Loops
- Bitwise Operations
- Shifting and Rotation

Activities:

Activity 1 (Loop and Array Processing Tasks):

1. Write a program to compute the factorial of 5 using a loop. Store result in memory.
2. Given two arrays A and B of 5 numbers each, write a program to find their intersection (common elements) and store them in array C.
   Example: A = [1, 2, 3, 4, 5], B = [3, 4, 6, 7, 8] → C = [3, 4].
3. Write a program to compute the difference of two arrays A – B (elements in A but not in B). Store the result in C.
   Example: A = [1, 2, 3, 4, 5], B = [3, 5, 6] → C = [1, 2, 4].

Activity 2 (Bit Manipulation and Shifting Tasks):

4. Calculate the number of one bits in BX and complement an equal number of least significant bits in AX. Hint: Use the XOR instruction and Rotate through carry.
   Example:
   BX: 1011 0001 1000 1001
   Total no of 1 Bitsin BX: 7
   Initial Value of AX: 1010 1011 1010 0101
   Value of AX after complementing: 1010 1011 1101 1010
5. Write a program that shifts a 64-bit number.
6. Write a program that adds two 64-bit numbers.
7. Write a program to multiply two 32-bit numbers and store the answer in a 64-bit location.
   dq 0xABCDD4E1 ; dq allocates 64 bit memory space. a is 32-bit number but it has space allocation of 64 bits
   dd 0xAB5C32 ; 32-bit space for multiplier
   result: dq 0x0 ; result should be 0x73005CB8FF6FF2 verify on calculator programmer's view
8. Write a program to check whether the given number is a power of two. If yes, store 1 in Result, else store 0.
   Example:
   Input: 8
   Binary: 1000
   Result: 1 (Yes, 8 is a power of 2)

   Input: 12
   Binary: 1100
   Output: 0 (No, 12 is not a power of 2)
9. Write a program that swaps the lower 4 bits (nibble) with the upper 4 bits of an 8-bit number.

Example:
Input: 0xAB = 1010 1011
Upper nibble = 1010 (A), Lower nibble = 1011 (B)
After Swap: 0xBA = 1011 1010

10. Write an Assembly Language program to check whether an 8-bit number is even or odd using bit manipulation.
    - If the number is even, store 1 in Result.
    - If the number is odd, store 0 in Result.
11. Write an Assembly Language program to invert the alternate bits bit 0, bit 2, bit 4, and bit 6 of an 8-bit register.
12. Write a program to count how many bits are set to 1 in a 16-bit number. Store the result in memory location BitCount.
    Example:
    Input: 0x6F2B = 0110 1111 0010 1011
    Output: BitCount = 10
13. Take a number in AX = 12.
    - Use SHL to multiply it by 2.
    - Use SHR to divide it by 2.

    Store results in memory.
14. Write a program to reverse the bits of a given byte.
    Example: 1101 0010b → 0100 1011b.
15. Given a byte in AL:
    - Clear the highest bit.
    - Toggle the third bit.
    - Check if the lowest bit is set (if yes, store 1 in result; else store 0).